PRE-ALGEBRA (7TH) · DATA ANALYSIS AND PROBABILITY

Spinner Probability Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Spinner Probability Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A spinner has 3 red parts, 2 blue parts and 1 green part, all the same size. What is P(red)?

2. If you spin 10 times, will you get exactly 5 reds? If you spin 1,000 times, about how many reds do you expect?

2 Explore use the tool

1. Clear the results and spin 10 times. Do this three times. Record the share of red each time.

2. Spin 1,000 or more times. How close is each color to its theoretical probability?

3. Build a spinner with P(red) = $\frac{1}{4}$ that uses 3 colors. List your parts.

4. Red came up 4 times in a row. Is blue more likely on the next spin? Explain.

3 Record use the tool and fill in the table

Theoretical and expected results

Parts are listed as Red, Blue, Green, Yellow.

Parts (R, B, G, Y)	P(red)	P(not red)	Expected reds in 300 spins
3, 2, 1, 0			
1, 1, 1, 1			
2, 5, 1, 0			
4, 1, 0, 1			
0, 3, 3, 2			

4 Explain what did you discover?

1. Theoretical probability = _____ ÷ _____.
2. Experimental probability = _____ ÷ _____.
3. As the number of spins grows, the experimental probability usually gets _____ to the theoretical probability.
4. Expected number of results = probability × _____.
5. Maya spun 20 times and got red 15 times, so she says P(red) = $\frac{3}{4}$ for a spinner with 3 red parts out of 6. Is she right? Explain.

5 Apply a new situation, no tool

1. A bag has 4 red, 6 blue and 10 green marbles. Find P(blue). If you draw (and replace) 50 times, about how many blues do you expect?

2. A number cube is rolled 600 times. About how many 5s do you expect?

3. A spinner with 4 equal parts (1 green) landed on green 18 times in 60 spins. Find the experimental P(green). Is the result reasonable?